

Problem Chosen

B

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Summary Sheet**

Team Control Number

2623614

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Summary

Keywords: keyword1; keyword2

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7 Conclusions

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1 Introduction

1.1 Background

Konstantin Tsiolkovsky 1895 “ ” 21 ———
MCM 2050 10 MCM

1.2 Literature Review

1.3 The Description of the Problem

1.4 Our work

2 Assumptions

Assumption 1: The light attenuation and scattering in the underwater environment adhere to the Jaffe-McGlamery imaging model, where the propagation of light follows an exponential decay law.

3 Notations

The core symbols and their definitions used in this study are summarized in Table 1, providing an overview of the key parameters and their related meanings.

Table 1: Notations used in this literature

Symbol	Description
x	Number of rocket launches per year
M	Total mass to be transported to space (kilogram)
T	Time to complete the mission (years)
C_{total}	Total cost of the mission (dollar)
C_{SES}	Cost per kilogram of space elevator system (dollar/kg)
C_{launch}	Cost per kilogram of rocket launch (dollar/kg)
C_{total_0}	Average total market transportation cost in the base year (current)
C_{fuel}	The theoretical fuel cost floor per 100kg of cargo
α	The annual technology iteration decay factor for manufacturing and operating costs, with a value of 0.05
P_{SES}	Annual throughput of the space elevator system (kilogram/year)
P_{launch}	Payload capacity per rocket launch (kilogram/launch)
PSNR	Peak signal-to-noise ratio for image quality evaluation
UCIQE	Underwater Color Image Quality Evaluation metric
UIQM	Underwater Image Quality Measure

4 Models

4.1 Modeling and Solving of Problem 1

4.1.1 Calculation of each scheme

Scheme a: Only Space Elevator System
ISEC $C_{SES} = \$100/\text{kg}$

$$17.9 \quad P_{SES} = 3 \times 179000 = 537000 \text{ kg/year}$$

$$T = \frac{M}{P_{SES}}$$

Total cost of the mission

$$C_{total} = C_{SES} \times M$$

$$186.22 \quad 10$$

Scheme b: Only Rocket Launches
\$500/kg 40

$$P_{launch} = 100000 \text{ kg}$$

$$C_{launch} =$$

$$x = \frac{M}{P_{launch} \times T}$$

Total cost of the mission

$$C_{total} = C_{launch} \times M$$

$$40 \quad 50 \quad 25000 \quad 68 \quad 7$$

Scheme c: Combined Space Elevator and Rocket Launches

$$T \quad C_{total}$$

x To analyze the trade-off between mission completion time and total cost in a hybrid space transportation system, a Hybrid Capacity Trade-off Optimization model with explicit Pareto Mapping (HCTO-PM) is developed.

$$x \quad T \quad C_{total}$$

$$T$$

$$T = \frac{M}{P_{SES} + P_{launch} \times x}$$

$$C_{total}$$

$$C_{total}(x) = C_{SES} \cdot P_{SES} \cdot \frac{M}{P_{SES} + x P_{launch}} + C_{launch} \cdot P_{launch} \cdot x$$

Multi-Objective Formulation: The model simultaneously considers the following two conflicting optimization objectives.

$$\begin{cases} \min T(x) = \frac{M}{P_{SES} + x P_{launch}} \\ \min C(x) = C_{total}(x) \end{cases}$$

Without the assumption of explicit preference weights, this problem does not pursue a single optimal solution, but instead constructs a Pareto optimal solution set to characterize the efficiency boundary in the feasible decision space. Since the model contains only one continuous decision variable, this paper uses the parameter sweep method to perform high-resolution sampling of the feasible interval of x . — Pareto

Table 2: Trade-off Results under Different Rocket Launch Frequencies

Rocket Launches x (times/year)	Completion Time T (years)	Total Cost C (million USD)
2,496	109.72	193,110.40
4,997	77.72	378,986.12
7,499	49.09	752,636.23

4.1.2 Improvement of scheme c

Rocket Iteration: Transportation cost iteration : 100kg — Learning
Curve Model Wright's Law T.P. Wright 1936
Dual-Factor Cost Structure
 C_{fuel} Wright's Law “ ” Learning by Doing C_{body}

$$C_{launch} = C_{man}(n) + C_{fuel} = C_1 \cdot n^{-b} + C_{fixed}$$

n b C_{fixed} “ ” “ ”
54,500 0.1% SpaceX 9 Falcon 9 Starship
ARK Invest SpaceX 9 20-30
30%-35% Learning Rate Horizontal Asymp-
tote

: “ ” “ ”
- α 5% - 100kg C_{fuel}
: 100kg $C_{launch}(t)$ t t ($t = 0$) $C_{launch}(t)$ $C_{man}(t)$ C_{fuel}

$$C_{launch}(t) = C_{man_0} \times (1 - \alpha)^t + C_{fuel}$$

$$C_{launch}(t) = (C_{total_0} - C_{fuel}) \times (0.95)^t + C_{fuel}$$

$$t (1 - \alpha)^t t C_{launch}(t) C_{fuel}$$

A new strategy based on the iterative models HCTO-PM t
Multi-Year Hybrid Capacity Trade-off Optimization with Pareto Mapping (MY-HCTO-PM) x t

$$T C_{total}$$

$$P_{SES}T + P_{launch} \cdot \sum_{t=0}^T x(t) = M$$

$$C_{total} = C_{SES} \cdot P_{SES} \cdot T + \sum_{t=0}^T C_{launch}(t) \cdot x(t) \cdot P_{launch}$$

T , (T, C) - Balanced
 (T_{min}, C_{min}) - Faster - Cheaper

4.1.3 Conclusions of Problem

MY-HCTO-PM

4.2 Modeling and Solving of Problem 2

4.2.1 Problem Analysis

4.2.2 Model Preparation

4.2.3 Model Construction

4.2.4 Model Solution

5 Sensitivity Analysis

6 Model Evaluation and Promotion

6.1 Model Evaluation

6.1.1 Advantages

6.1.2 Limitations

6.2 Future Work

6.2.1 Model extension

6.2.2 Model application

7 Conclusions

References

- [1] L. Prandtl, Fluid motions with very small friction, Proceedings of the 3rd International Mathematical Congress, Heidelberg: H. Schlichting, 1904, 484-491.

Appendices

Report on Use of AI

1. Bing AI

Query1: write a sort algorithm

Output: Sure, I can help you with that. Here is a simple implementation of the selection sort algorithm in Python:

```
def selection_sort(arr):
    for i in range(len(arr)):
        min_idx = i
        for j in range(i + 1, len(arr)):
            if arr[min_idx] > arr[j]:
                min_idx = j
        arr[i], arr[min_idx] = arr[min_idx], arr[i]
    return arr
```

This algorithm sorts an array by repeatedly finding the minimum element from the unsorted part of the array and putting it at the beginning. The time complexity of this algorithm is $O(n^2)$.

There are many other sorting algorithms available such as bubble sort, insertion sort, merge sort, quick sort, heap sort, and more. Each of these algorithms has its own advantages and disadvantages. You can learn more about them on [GeeksforGeeks](#) and [Programiz](#).

I hope this helps!

2. OpenAI Enie (Nov 5, 2023 version, Ernie 4.0)

Query1: <insert the exact wording of any subsequent input into the AI tool>

Output: <insert the complete output from the second query>

3. Github CoPilot (Feb 3, 2024 version)

Query1: <insert the exact wording you input into the AI tool>

Output: <insert the complete output from the AI tool>

4. Google Bard (Feb 2, 2024 version)

Query: <insert the exact wording of your query>

Output: <insert the complete output from the AI tool>

5. Bing AI

Query: write a FFT algorithm

Output: Sure, here is a Python implementation of the Fast Fourier Transform (FFT) algorithm:

```
import numpy as np

def fft(x):
    n = len(x)
    if n == 1:
        return x
    even = fft(x[0::2])
    odd = fft(x[1::2])
    T = [np.exp(-2j * np.pi * k / n) * odd[k] for k in range(n // 2)]
    return [even[k] + T[k] for k in range(n // 2)] + [even[k] - T[k] for k
        ↪ in range(n // 2)]
```

The FFT is a widely used algorithm for computing the **Discrete Fourier Transform (DFT)** of a sequence. It reduces the number of computations needed from $O(n^2)$ to $O(n \log n)$ 12. The basic idea of the FFT is to apply divide and conquer. We divide the coefficient vector of the polynomial into two vectors, recursively compute the DFT for each of them, and combine the results to compute the DFT of the complete polynomial 3.

I hope this helps!